



Evaluating Multi-Cycle Fiber-Reinforced Ceramic Matrix Composites

Bailey Walke, Dr. Gurpreet Singh

Department of Mechanical & Nuclear Engineering, Kansas State University, Manhattan, Kansas, USA



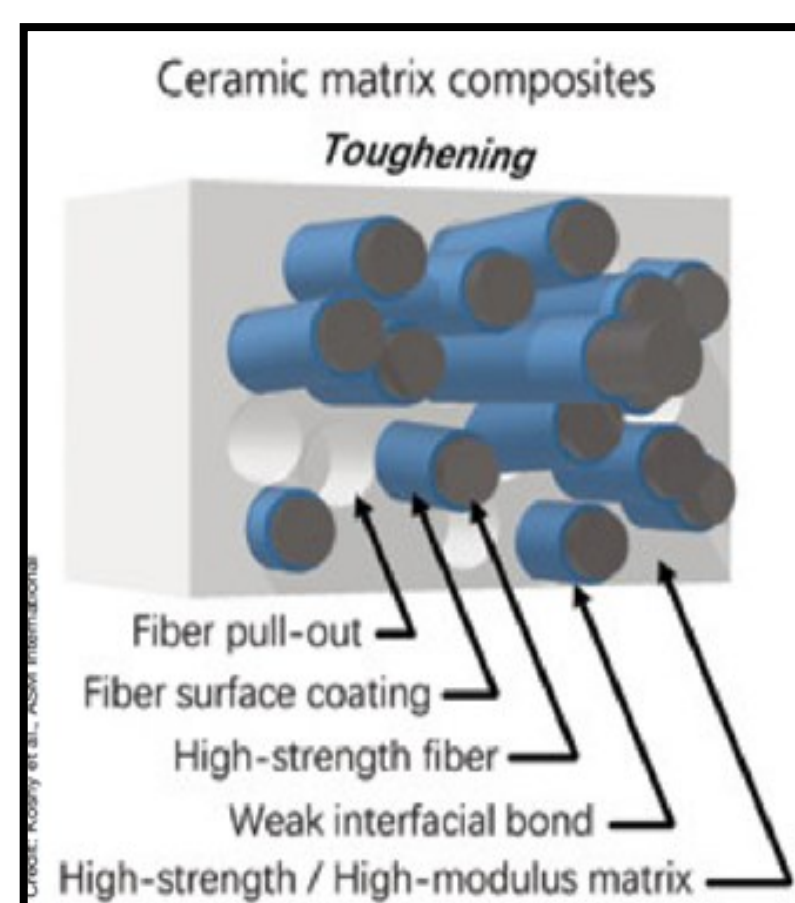
Project Objectives and Goals

This project explores the capabilities of multi-cycle Ceramic Matrix Composites (CMCs). Data collection of chemical, physical and visual properties of these samples will be used to determine if multiple cycles improves the quality of the CMC.

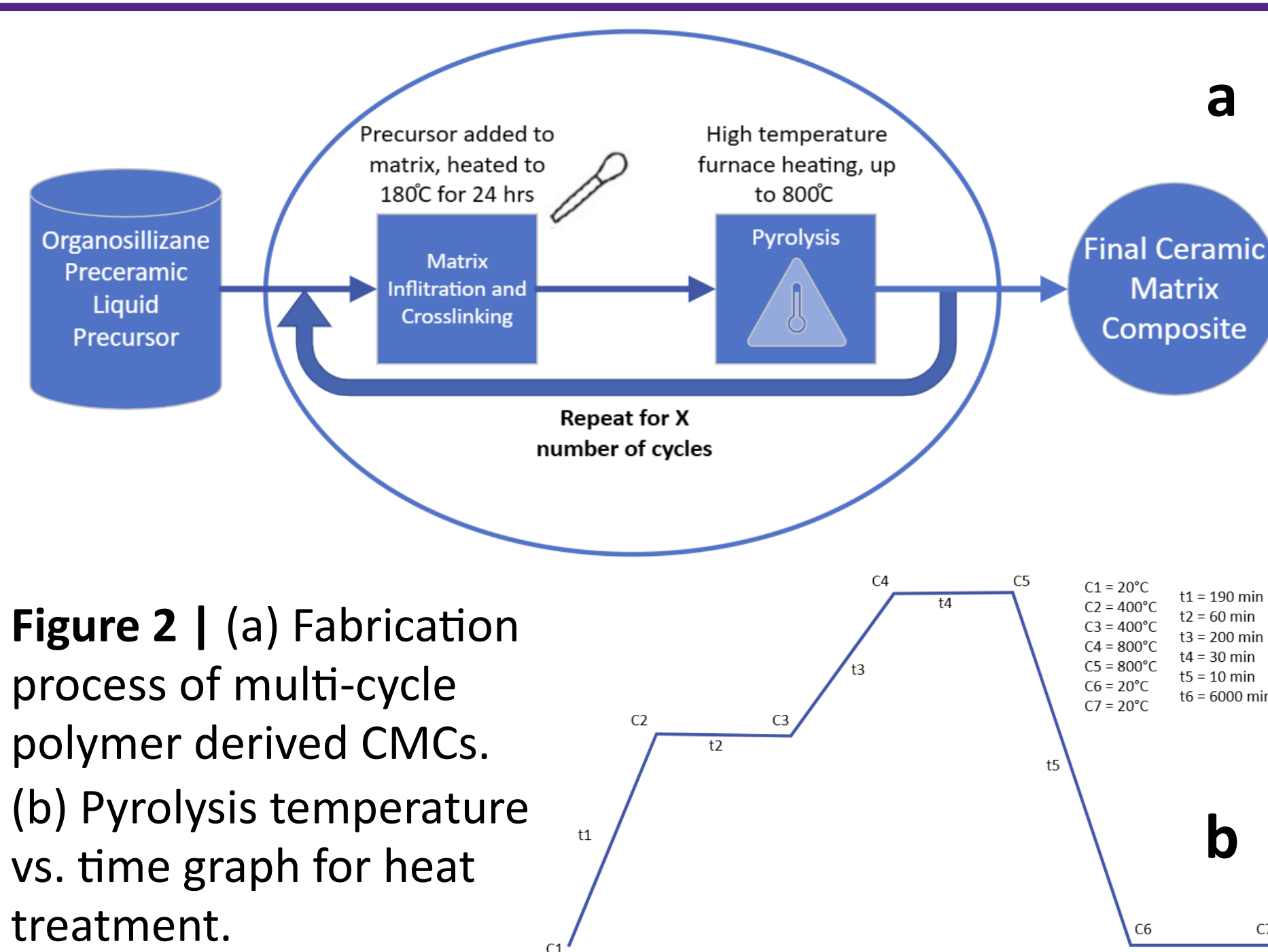
Background

- Aerospace applications such as aircraft engines and rocket nozzles demand high-performing materials in extreme environments.
- CMCs are low density, damage tolerant, and oxidation resistant composite materials. They retain their material and mechanical properties in extreme environments and high temperatures.

Figure 1 | Desired composition of CMCs. Ceramic matrix is shown in light grey and carbon fiber reinforcement is shown in dark grey. [2]



Experiment Setup



Data & Results

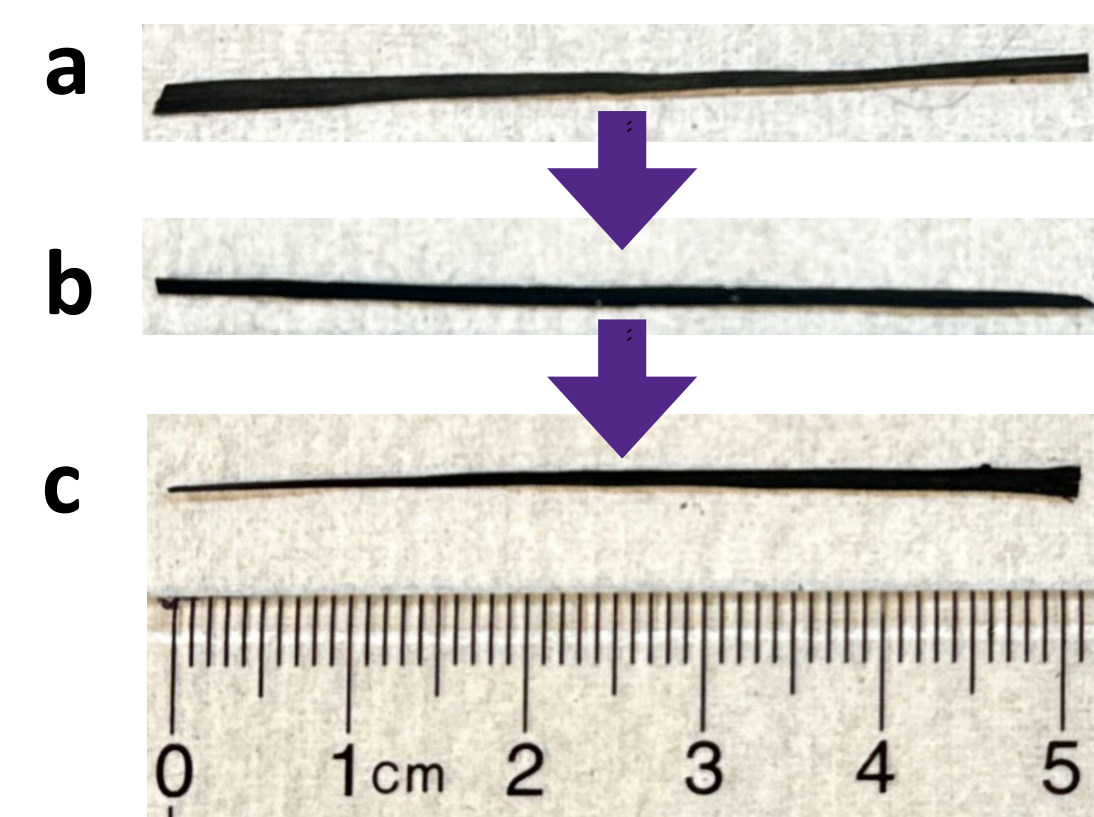


Figure 3 | Neat carbon fiber (a), crosslinked (b), and pyrolyzed (c) samples. 5cm in length.

Type	Cycle	Pyrolyzed Yield	Ceramic Yield
SiCN	1	86.62%	60.27%
	2	89.12%	36.37%
	3	92.51%	26.69%
SiBCN	1	96.36%	23.56%
	2	94.63%	18.85%
	3	96.46%	11.71%

Table 1 | SiCN and Si(B)CN averaged pyrolyzed and ceramic yields per cycle.

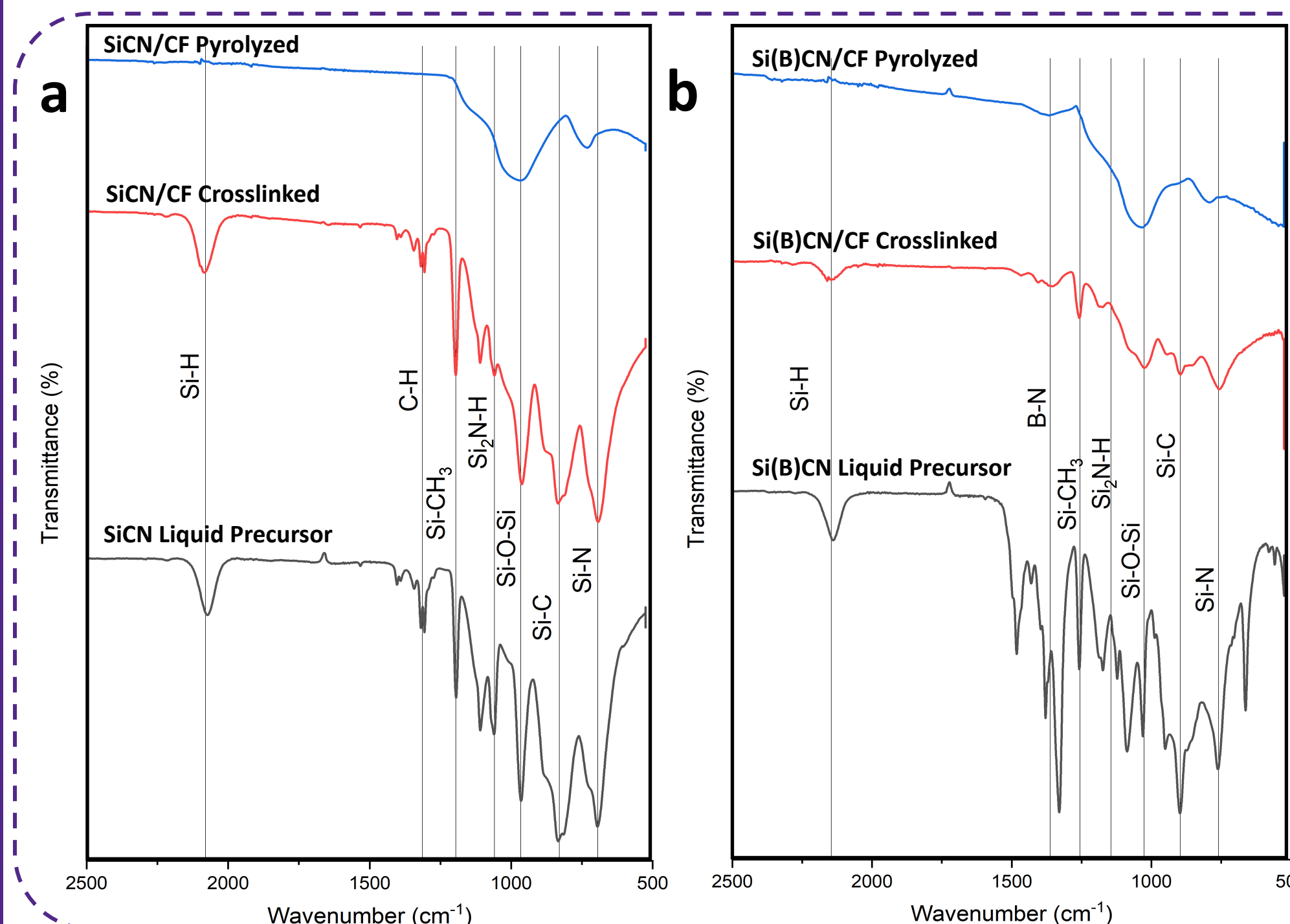


Figure 4 | FTIR spectroscopy of liquid precursor, cross-linked, and pyrolyzed samples. (a) SiCN/SiC samples FTIR. (b) Si(B)CN/SiC sample FTIR.

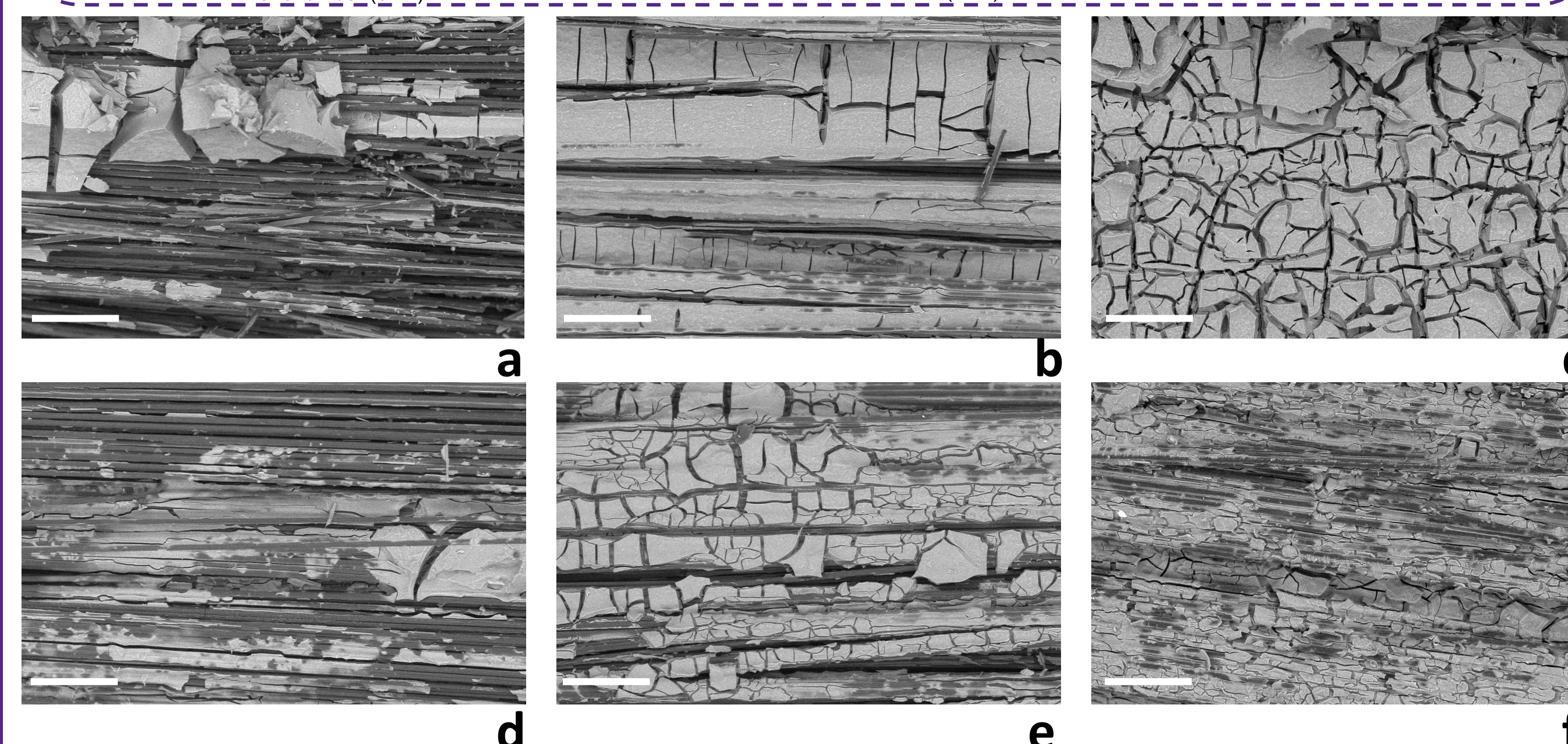


Figure 5 | SEM images at 800x magnification of SiCN-CF Cycle 1(a), Cycle 2(b), Cycle 3(c) and Si(B)CN-CF Cycle 1(d), Cycle 2(e), Cycle 3(f). Scale bar is 100µm.

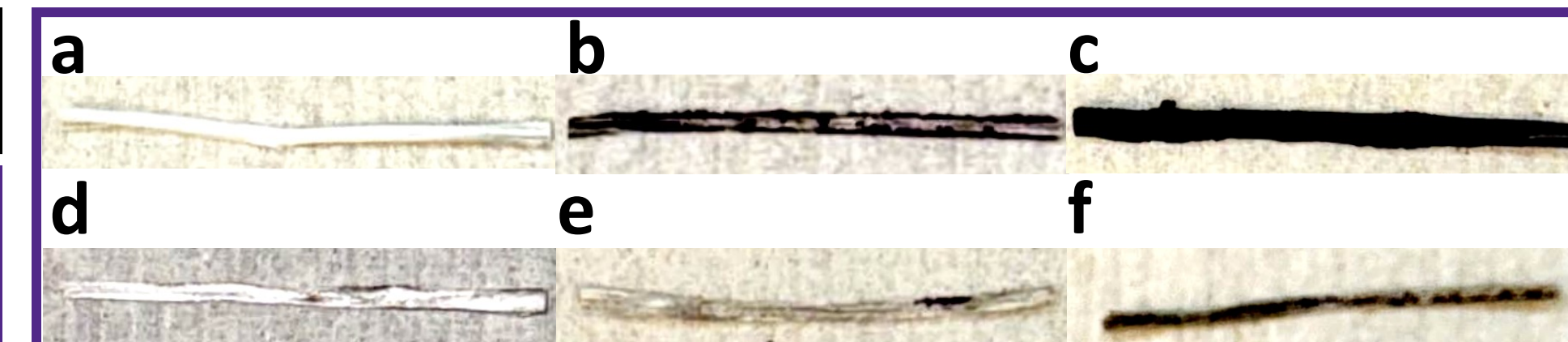


Figure 6 | Oxidation results of SiCN-CF Cycle 1(a), Cycle 2(b), Cycle 3(c) and Si(B)CN-CF Cycle 1(d), Cycle 2(e), Cycle 3(f). Samples were tested at 800°C.

Type	Cycle	Weight retention (%) at 800°C
SiCN	1	27.03%
	2	48.21%
	3	69.71%
SiBCN	1	18.33%
	2	29.33%
	3	36.05%

Table 2 | Oxidation weight retention of SiCN and Si(B)CN cycles.

Conclusions

- FTIR spectroscopy peaks show that correct bonds are formed for both precursor types.
- SEM imaging confirms the formation of a ceramic matrix on the carbon fiber reinforcement.
- Oxidation testing shows that higher cycle CMCs have improved weight retention.

Future Studies

- Tensile testing of bundle samples to determine elastic modulus.
- Using SEM images to measure the changes in fiber diameter between each cycle.
- Oxidation testing at higher temperatures.

References

- Mujib, S. (2022). Evaluating Use of Boron- and Hafnium-Modified Polysilazanes for Ceramic Matrix Minicomposites. Kansas State University
- Lanser, R.L. (2015). Tension-Compression Fatigue of an Oxide/Oxide Ceramic Matrix Composite at Elevated Temperature in Air and Steam Environments.